# BLOW-UP SOLUTIONS FOR MEAN FIELD EQUATIONS WITH NEUMANN BOUNDARY CONDITIONS ON RIEMANN SURFACES

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## Abstract

On a compact Riemann surface $(\Sigma, g)$ with smooth boundary $\partial\Sigma$, we consider the following mean field equations with Neumann boundary conditions:

$$\begin{cases} -\Delta_g u = \lambda \left( \dfrac{Ve^u}{\int_\Sigma Ve^u dv_g} - \dfrac{1}{|\Sigma|_g} \right) & \text{in } \mathring{\Sigma} \\ \partial_\nu u = 0 & \text{on } \partial\Sigma \end{cases} . \tag{1}$$

We proved the existence of solutions that blow up as $\lambda$ approaches a critical parameter

$$\lambda_{km} := 8k\pi + 4(m-k)\pi, k \le m \in \mathbb{N}_+.$$

The blow-up may occur at a prescribed number of points in the interior $\mathring{\Sigma}$ and on the boundary $\partial\Sigma$. To derive the blow-up solutions, we use the Lyapunov-Schmidt reduction and some variational techniques.

**Key Words:** Mean field equations; Exponential non-linearity; Blow-up solutions; Lyapunov-Schmidt reduction

## Introduction

***Background:*** Mean field equations arise in many applications, including problems such as the Kazdan-Warner problem, prescribed Gauss curvature problem, the Chern-Simons-Higgs gauge theory, the Keller-Segel system for chemotaxis collapse, statistic mechanics and others.

**Motivation:** One can prove the existence of solutions on a compact Riemann surface $\Sigma$ with boundaries for the parameter $\lambda \notin 4\pi\mathbb{N}_+$, while when $\lambda$ approaches to the critical parameter value $4k\pi$, the blow-up phenomenon may occur. Historically, most research has focus on mean field equation equipped with Dirichlet boundary conditions.

- Esposite-Grossi-Pistoia in [2] found that the "stable" critical points of a special function $\mathcal{F}$ involved with $V$ and Green's function can be the blow-up points of a sequence of blow-up solutions as $\lambda \to 8m\pi$ on domains;
- Esposito-Figueroa in [1] extended the result to singualr mean field equation on a closed surface.

**Goal:** We aim to construct solutions for the mean field equations (1) on a compact Riemann surface $\Sigma$ with smooth boundary $\partial\Sigma$ near critical parameter values, where the solutions may blow up as $\lambda$ approaches $4k\pi$.

## Discussion

**Related problems:**

- Singular mean field equations
  $-\Delta_g u = \lambda \left( \frac{Ve^u}{\int_\Sigma Ve^u dv_g} - \frac{1}{|\Sigma|} \right) + \frac{4\pi \sum_i n_i}{|\Sigma|} - 4\pi \sum_{j=1}^{l} n_j \delta_{p_j}$;
- The steady state of the Keller-Segel system:
  $-\Delta_g u + \beta u = \lambda \left( \frac{Ve^u}{\int_\Sigma Ve^u dv_g} - \frac{1}{|\Sigma|_g} \right)$;
- Toda system: $\begin{cases} -\Delta_g u_1 = 2\rho_1 \left( \frac{V_1 e^{u_1}}{\int_\Sigma V_1 e^{u_1} dv_g} - 1 \right) - \rho_2 \left( \frac{V_2 e^{u_2}}{\int_\Sigma V_2 e^{u_2} dv_g} - 1 \right) \\ -\Delta_g u_2 = 2\rho_2 \left( \frac{V_2 e^{u_2}}{\int_\Sigma V_2 e^{u_2} dv_g} - 1 \right) - \rho_1 \left( \frac{V_1 e^{u_1}}{\int_\Sigma V_1 e^{u_1} dv_g} - 1 \right) \end{cases}$ .

## References

[1] P. Esposito and P. Figueroa.
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## Method

**Energy functional:** $J_\lambda(u) := \frac{1}{2}\int_\Sigma |\nabla u|^2 dv_g - \lambda \log\left(\int_\Sigma Ve^u dv_g\right)$, where $u \in \overline{\mathrm{H}}^1(\Sigma) := \left\{ u \in H^1(\Sigma) : \int_\Sigma u dv_g = 0 \right\}$.

**Approximation Solutions:** By isothermal coordinate $(y_\xi, U(\xi))$, we define $U_{\tau,\xi}(x) = \log \frac{8\tau^2}{(\tau^2\rho^2 + |y_\xi(x)|^2)}$, for all $x \in U(\xi)$. And $PU_{\tau,\xi}$ is a projection of $U_{\tau,\xi}$ into space $\overline{\mathrm{H}}^1(\Sigma)$ by

$$\begin{cases} -\Delta_g PU_{\tau,\xi}(x) = \rho^2 \chi_\xi(x) e^{-\varphi_\xi(x)} e^{U_{\tau,\xi}(x)} - \overline{\rho^2 \chi_\xi(x) e^{-\varphi_\xi(x)} e^{U_{\tau,\xi}(x)}}, & x \in \mathring{\Sigma} \\ \partial_\nu PU_{\tau,\xi} = 0, & x \in \partial\Sigma \\ \int_\Sigma PU_{\tau,\xi} dv_g = 0 \end{cases} ,$$

where $\chi_\xi$ is a cut-off function around $\xi$.

For any $\xi = (\xi_1, \cdots, \xi_m) \in\in \mathring{\Sigma}^k \times (\partial\Sigma)^{m-k} \setminus \Delta$, we choose $\tau_i(\xi) = \sqrt{\frac{1}{8} V(\xi_i) e^{\sigma(\xi_i) H(\xi_i,\xi_i) + \sum_{h \ne i} \sigma(\xi_h) G(\xi_i,\xi_h)}}$. Setting $PU_i = PU_{\tau_i(\xi),\xi_i}$, the functions in the following manifold:

$$\mathcal{M}_\rho^m := \left\{ \sum_{i=1}^m PU_i : \xi \in \mathring{\Sigma}^k \times (\partial\Sigma)^{m-k} \right\}$$

are good approximation solutions of the equations (1).

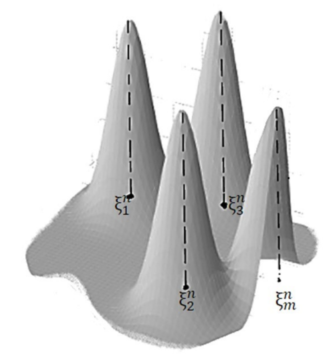

Figure 1: Schematic of the blow up solution

The solution $u$ can decompose into two parts: $u = \sum_{i=1}^m PU_i + \phi_\xi^\rho$, where $\phi_\xi^\rho$ is the remainder term.

**The Lyapunov-Schmidt reduction:**

- For any $\xi$ in compact subset of $\mathring{\Sigma}^k \times (\partial\Sigma)^{m-k} \setminus \Delta$, there is a unique $\phi_\xi^\rho$ such that $\sum_i PU_i + \phi_\xi^\rho$ solves (1) in the orthogonal complement of $T_\xi \mathcal{M}_\rho^m$ with $\|\phi_\xi^\rho\| = o(\rho^{\alpha_0} |\log\rho|)(\rho \to 0)$, for some $\alpha_0 \in (0,1)$.

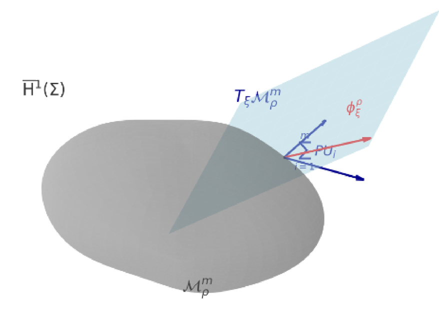

Figure 2: The Lyapunov-Schmidt reduction

- The function $\sum_{i=1}^m PU_{\tau(\xi),\xi_i} + \phi_\xi^\rho$ is a solution of (1) iff $\xi$ is a critical point of the reduced functional $\qquad \tilde{E}_\rho(\xi) = E_\rho\left(\sum_{i=1}^m PU_{\tau_i(\xi),\xi_i} + \phi_\xi^\rho\right)$,
  where $E_\rho(u) = \frac{1}{2}\int_\Sigma |\nabla u|_g^2 dv_g - \rho^2 \int_\Sigma Ve^u dv_g$.

**The expansion of the reduced functional:**

$$\widetilde{E}_\rho(\xi) = \sum_{i=1}^m \sigma(\xi_i)(3\log 2 - 2\log\rho) - 2\sum_{i=1}^m \sigma(\xi_i) - \frac{1}{2}\mathcal{F}(\xi) + o(1),$$

as $\rho \to 0$, $C^1$-uniformly in any compact set of $\mathring{\Sigma} \times (\partial\Sigma)^{m-k} \setminus \Delta$.

**Proof of the Main Results:**

- Theorem 1 follows from the Lyapunov-Schmidt reduction and expansion of the reduced functional;
- The statement $\mathcal{F}(\xi) \to +\infty$, as $\xi$ going to $\partial(\mathring{\Sigma}^k \times (\partial\Sigma)^{m-k} \setminus \Delta))$ implies that there exists a minimum point of $\mathcal{F}$ which is stable. Thus the equations (1) admit a sequence of blow up solutions along $\lambda_n \to \lambda_{km}$.

## Main Results

**Theorem 1.** Fix $k \le m \in \mathbb{N}_+$ and let $\sigma(\xi)$ be defined as: $8\pi$ if $\xi \in \mathring{\Sigma}$; $4\pi$ if $\xi \in \partial\Sigma$. $\Delta := \{\xi \in \Sigma^m : \xi_i = \xi_j \text{ for some } i \ne j\}$. If $K \subset\subset \mathring{\Sigma}^k \times (\partial\Sigma)^{m-k} \setminus \Delta$ is a $C^1$-stable critical point set of the function

$$\mathcal{F} : \mathring{\Sigma}^k \times (\partial\Sigma)^{m-k} \setminus \Delta \to \mathbb{R},$$

$$\mathcal{F}(\xi) = \sum_{i=1}^m \sigma^2(\xi_i) H(\xi_i,\xi_i) + \sum_{\substack{i,j=1 \\ i \ne j}}^m \sigma(\xi_i)\sigma(\xi_j) G(\xi_i,\xi_j) + \sum_{i=1}^m 2\sigma(\xi_i) \log V(\xi_i).$$

Then for any $n \in \mathbb{N}_+$ there exists a parameter $\lambda_n$ in a small neighborhood of $\lambda_{km}$ satisfying that $\lambda_n \to \lambda_{km} (n \to +\infty)$ and $u_n$ which solves problem (1) with respect to parameter $\lambda_n$ blows up precisely at points $\xi_1, \ldots, \xi_m$ with the property $(\xi_1, \ldots, \xi_m) \in K$ (up to a subsequence), in a sense that

$$\frac{\lambda_n e^{u_n}}{\int_\Sigma e^{u_n}} \rightharpoonup \sum_{i=1}^m \sigma(\xi_i)\delta_{\xi_i}, \text{ as } n \to +\infty$$

which is weakly convergent as measures in $\Sigma$.

**Theorem 2.** With assumptions above and $\inf_\Sigma V > 0$, there exists a sequence of blow up solutions which blow up at points $\xi_1, \cdots, \xi_m$ with $(\xi_1, \cdots, \xi_m)$ as the minimum point of $\mathcal{F}$ on $\mathring{\Sigma}^k \times (\partial\Sigma)^{m-k}$ along $\lambda \to \lambda_{km}$.